

Market challenges

- Inadequate production infrastructure for low-cost/cost-effective products
- Pressure from cost-sensitive markets
- Limited availability of high-quality products
- Limited scale of electronics manufacturing in the country
- High import costs
- Unfavourable tax and duty structures

TABLE I
CLASSIFICATION OF ELECTRONICS CHEMICALS AND MATERIALS

By type of products	By application	By form
1. Silicon wafers	1. Semiconductor packaging	1. Solid
2. PCB laminates	2. Integrated circuits manufacturing	2. Liquid
3. Specialty gases	3. PCB manufacturing	3. Gaseous
4. Photoresists		
5. Wet chemicals		
6. CMP slurries		
7. Low-K dielectrics		
8. Others		

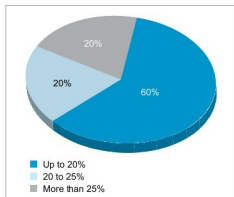


Fig. 2: Expected growth rate of the Indian market for electronics chemicals and materials

The wafer fabrication segment accounted for the major share of revenue in 2015, and is expected to maintain its lead throughout the analysis period. Silicon is the most common type of wafer fabrication material used globally, and accounts for about 17 per cent of the market share.

The same report states that demand for electronics chemicals and materials has increased owing to technological advancements as well as an increase in demand from applications like photovoltaics, integrated circuits, LED bulbs and PCBs for electronic gadgets such as mobiles, laptops, computers and tablets.

There is a shift towards advanced technology nodes, which is driving the market for wafer fabrication chemicals and materials. Other factors that are driving this market's growth include environmental regulations, an increase in consumption of integrated circuit process chemicals, technological innovations, increasing demand for electronics and rapid economic growth in emerging economies.

Sixty per cent of the survey participants forecast that the Indian market for chemicals and materials used in the electronics manufacturing industry will grow at a rate of up to 20 per cent. However, 20 per cent of the survey participants predict a growth rate of between 20 to 25 per cent. Remaining 20 per cent expect an even more accelerated growth, which is closer to 25 per cent.

Market mix

Materials used in electronics manufacturing are well-known, high-purity chemicals such as photoresists, wet chemicals, acids, gases and solvents, and electronic materials such as wafers and laminates. Based on the types of products used, electronics materials and chemicals market can be classified into eight segments—silicon wafers, PCB laminates, photoresists, specialty gases, CMP slurries, wet chemicals, low-K dielectrics and others (Table I).

Demand side

Silicon wafers constitute the major global market in electronics chemi-

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